

CO4 – AT2

INDUSTRY-BASED PROBLEM SOLVING

NAME: V.SAI CHARAN

REG NO: 192472159

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COURSE NAME: FUNDAMENTALS OF DATA SCIENCE

Question 11 – Digital Marketing

Industry Area	Digital Marketing
Topic	Hypothesis Testing Using Confidence Interval
Problem Statement	Before introducing a new advertisement, the conversion rate was 8%. After the campaign, 90 out of 1,000 visitors converted. Using a 95% confidence interval, determine whether the observed conversion rate provides evidence of improvement.

1. Population and Sample

Population: All visitors exposed to the new advertisement. Sample: 1,000 visitors observed after the campaign, of whom 90 converted.

2. Parameter of Interest

The population conversion proportion p after the campaign.

3. Point Estimate

Sample conversion rate $\hat{p} = x/n = 90/1000 = 0.090 = 9.0\%$.

4. Null Hypothesis (H_0)

$H_0: p = 0.08$. The post-campaign conversion rate is 8%.

5. Alternative Hypothesis (H_1)

$H_1: p > 0.08$. The campaign improves the conversion rate.

6. Significance Level

For a 95% confidence interval, $\alpha = 0.05$.

7. Confidence Interval

$SE = \sqrt{[\hat{p}(1-\hat{p})/n]} = 0.00905$. Margin of error $= 1.96 \times SE = 0.01774$. Therefore, the 95% CI is (0.0723, 0.1077), or (7.23%, 10.77%).

8. Test/Interval Interpretation

The previous conversion rate of 8% (0.08) lies inside the 95% confidence interval. Therefore, an 8% population conversion rate remains plausible after the campaign.

9. Decision

Because 0.08 lies within the 95% CI, reject H_0 is not justified at the 5% level. A two-sided check gives $z = 1.166$, $p \approx 0.2438$.

10. Industry-Oriented Conclusion

The campaign produced a sample conversion rate of 9%, but the increase from 8% is not statistically supported by the 95% confidence interval. The marketing team should not conclude that the advertisement definitively improved conversion performance based on this sample alone.

Question 12 – Education Technology

Industry Area	Education Technology
Topic	Hypothesis Testing Using Confidence Interval
Problem Statement	An EdTech company claims that its adaptive-learning platform improves the average test score to at least 75. A sample of 64 students obtains an average score of 78 with standard deviation 10. Use a 95% confidence interval to assess the claim.

1. Population and Sample

Population: Students using the adaptive-learning platform. Sample: 64 students whose average score is 78 with standard deviation 10.

2. Parameter of Interest

The population mean test score μ .

3. Point Estimate

The sample mean $\bar{x} = 78$ is the point estimate of μ .

4. Null Hypothesis (H_0)

For assessing the stated threshold, $H_0: \mu = 75$ (benchmark value).

5. Alternative Hypothesis (H_1)

$H_1: \mu > 75$, representing support for an average score above the benchmark.

6. Significance Level

$\alpha = 0.05$, corresponding to a 95% confidence interval.

7. Confidence Interval

Since the population standard deviation is not given, use a t interval with $df = 63$. $t_{0.975,63} = 1.998$. $SE = 10/\sqrt{64} = 1.250$. Margin of error = 2.498. 95% CI = (75.502, 80.498).

8. Test/Interval Interpretation

The entire 95% confidence interval, approximately 75.50 to 80.50, is above 75. This indicates that the population mean is plausibly greater than the claimed minimum benchmark.

9. Decision

The benchmark value 75 is below the lower confidence limit. Therefore, the evidence supports rejecting the benchmark equality in favour of $\mu > 75$ at the 5% level.

10. Industry-Oriented Conclusion

The sample provides statistically meaningful evidence that the adaptive-learning platform is associated with an average test score above 75. The EdTech company can reasonably report support for its performance claim, subject to the representativeness of the student sample.

Question 13 – Automobile Industry

Industry Area	Automobile Industry
Topic	Hypothesis Testing Using p-value
Problem Statement	A car manufacturer claims that its new model provides an average mileage of 20 km/litre. A sample of 50 cars gives a mean mileage of 19.2 km/litre with standard deviation 2.5 km/litre. Perform a hypothesis test at the 5% significance level using the p-value approach.

1. Population and Sample

Population: All cars of the new model under comparable operating conditions. Sample: 50 cars with mean mileage 19.2 km/litre and standard deviation 2.5 km/litre.

2. Parameter of Interest

The population mean mileage μ .

3. Point Estimate

$\bar{x} = 19.2$ km/litre is the point estimate of μ .

4. Null Hypothesis (H_0)

$H_0: \mu = 20$ km/litre.

5. Alternative Hypothesis (H_1)

$H_1: \mu \neq 20$ km/litre, because the claim is about equality to 20 km/litre.

6. Significance Level

$\alpha = 0.05$.

7. Test Statistic

Use a one-sample t-test because σ is unknown. $t = (\bar{x} - \mu_0) / (s / \sqrt{n}) = (19.2 - 20) / (2.5 / \sqrt{50}) = -2.263$, with $df = 49$.

8. p-value

Two-tailed p-value = 0.0281 (approximately).

9. Compare p-value with α

p-value (0.0281) < α (0.05). Therefore, reject H_0 .

10. Industry-Oriented Conclusion

There is statistically significant evidence that the true average mileage differs from the claimed 20 km/litre. Because the sample mean is only 19.2 km/litre, the result indicates that the manufacturer's 20 km/litre claim is not supported by the observed sample and warrants further performance investigation.

Question 14 – Cloud Computing

Industry Area	Cloud Computing
Topic	Hypothesis Testing Using p-value
Problem Statement	A cloud provider claims that its average server response time is 100 ms. A sample of 81 requests has a mean of 104 ms and standard deviation of 12 ms. Test the claim at $\alpha = 0.05$ using the p-value method.

1. Population and Sample

Population: All comparable server requests handled by the cloud service. Sample: 81 requests with mean response time 104 ms and standard deviation 12 ms.

2. Parameter of Interest

The population mean server response time μ .

3. Point Estimate

$\bar{x} = 104$ ms is the point estimate of μ .

4. Null Hypothesis (H_0)

$H_0: \mu = 100$ ms.

5. Alternative Hypothesis (H_1)

$H_1: \mu \neq 100$ ms.

6. Significance Level

$\alpha = 0.05$.

7. Test Statistic

Use a one-sample t-test because the population standard deviation is unknown. $t = (104 - 100) / (12 / \sqrt{81}) = 3.000$, with $df = 80$.

8. p-value

Two-tailed p-value = 0.00360 (approximately).

9. Compare p-value with α

p-value (0.00360) < α (0.05). Therefore, reject H_0 .

10. Industry-Oriented Conclusion

The sample provides strong statistical evidence that the cloud provider's average server response time is different from 100 ms. Since the observed mean is 104 ms, the evidence points toward slower average response than the claimed 100 ms benchmark. The provider should investigate workload, infrastructure, latency, and service configuration before treating 100 ms as the operational average.

Question 15 – E-commerce

Industry Area	E-commerce
Topic	Hypothesis Testing Using p-value
Problem Statement	An online company claims that its product return rate is 10%. In a sample of 1,000 orders, 130 products were returned. Test whether the actual return rate is significantly higher than 10% using the p-value approach at the 5% significance level.

1. Population and Sample

Population: All orders placed through the online company. Sample: 1,000 orders, of which 130 were returned.

2. Parameter of Interest

The population product return proportion p .

3. Point Estimate

$$\hat{p} = 130/1000 = 0.130 = 13.0\%.$$

4. Null Hypothesis (H_0)

$$H_0: p = 0.10.$$

5. Alternative Hypothesis (H_1)

$H_1: p > 0.10$, because the question asks whether the actual return rate is significantly higher.

6. Significance Level

$$\alpha = 0.05.$$

7. Test Statistic

Using the one-proportion z-test, $z = (\hat{p} - p_0) / \sqrt{[p_0(1 - p_0)/n]} = 3.162$.

8. p-value

Upper-tail p-value = 0.000783 (approximately).

9. Compare p-value with α

p-value (0.000783) < α (0.05). Therefore, reject H_0 .

10. Industry-Oriented Conclusion

There is statistically significant evidence that the actual product return rate is higher than 10%. The observed return rate is 13%, which is 3 percentage points above the claimed rate. The e-commerce company should examine product quality, sizing/description accuracy, delivery damage, customer expectations, and return reasons to identify the operational causes of the higher return rate.

Assessment Summary

Q.No.	Industry	Method	Key Result	Decision
11	Digital Marketing	95% CI for proportion	CI $\approx$ 7.23%–10.77%	No significant evidence of improvement
12	Education Technology	95% CI for mean	CI $\approx$ 75.50–80.50	Supports mean score $>$ 75
13	Automobile Industry	Two-tailed t-test	$t = -2.263$; $p \approx 0.0279$	Reject H_0
14	Cloud Computing	Two-tailed t-test	$t = 3.000$; $p \approx 0.0038$	Reject H_0
15	E-commerce	One-tailed proportion z-test	$z = 3.162$; $p \approx 0.0008$	Reject H_0 ; return rate is higher

Overall Assessment Note: The calculations above follow the problem statements and the ten required assessment points specified in the supplied source document.